Condition for Theorem 3.2

When $\frac{n_1}{N_1} \geq \frac{n_2}{N_2}$ and $k = j$, then

$$g_{n_1,k} \circ g_{n_2,j}^{-1}(t) - g_{N_1,k} \circ g_{N_2,j}^{-1}(t)$$

is non-negative when $t \in (0,1)$, where $g_{n,i}(x) = I_x(i, n-i+1)$, $I_x(\alpha, \beta)$ is the CDF of Beta(α, β) distribution.

We hope to find a sufficient condition of $g_{n_1,k} \circ g_{n_2,j}^{-1}(t) - g_{N_1,k} \circ g_{N_2,j}^{-1}(t)$ is non-negative when $t \in (0,1)$.

Define the function $f(x) = g_{n_1,k}(x) - g_{N_1,k}(x), x \in (0,1), n_1 \geq N_1$ and we will prove $f(x) \geq 0$ for all $x \in (0,1)$

Proof. we find $f(0) = 0$ and $f(1) = 0$, then it suffices to show $f^{'}(x)$ is positive first and then becomes negative.

$$\begin{aligned} f^{'}(x) &= \frac{x^{k-1}(1-x)^{n_1-k}}{B(k, n_1+1-k)} - \frac{x^{k-1}(1-x)^{N_1-k}}{B(k, N_1+1-k)} \\ &= \frac{x^{k-1}(1-x)^{N_1-k}}{B(k, n_1+1-k)}((1-x)^{n_1-N_1} - \frac{B(k, n_1+1-k)}{B(k, N_1+1-k)}) \end{aligned}$$

When $x = 0$, $(1-x)^{n_1-N_1} = 1$ and when $x = 1$, $(1-x)^{n_1-N_1} = 0$. Also $\frac{B(k,n_1+1-k)}{B(k,N_1+1-k)} \leq 1$.

Then $f^{'}(x)$ is positive first and then becomes negative. And now we have $f(x) \geq 0$ for all $x \in (0,1)$.

Then when $x = g_{n_2,j}^{-1}(t)$, where $t \in (0,1)$, we have

$$g_{n_1,k}(g_{n_2,j}^{-1}(t)) \geq g_{N_1,k}(g_{n_2,j}^{-1}(t)) \tag{1}$$

Since $g_{N_1,k}(x)$ is an increasing function and $g_{n_2,j}^{-1}(t) \geq g_{N_2,j}^{-1}(t)$ when $n_2 \leq N_2$

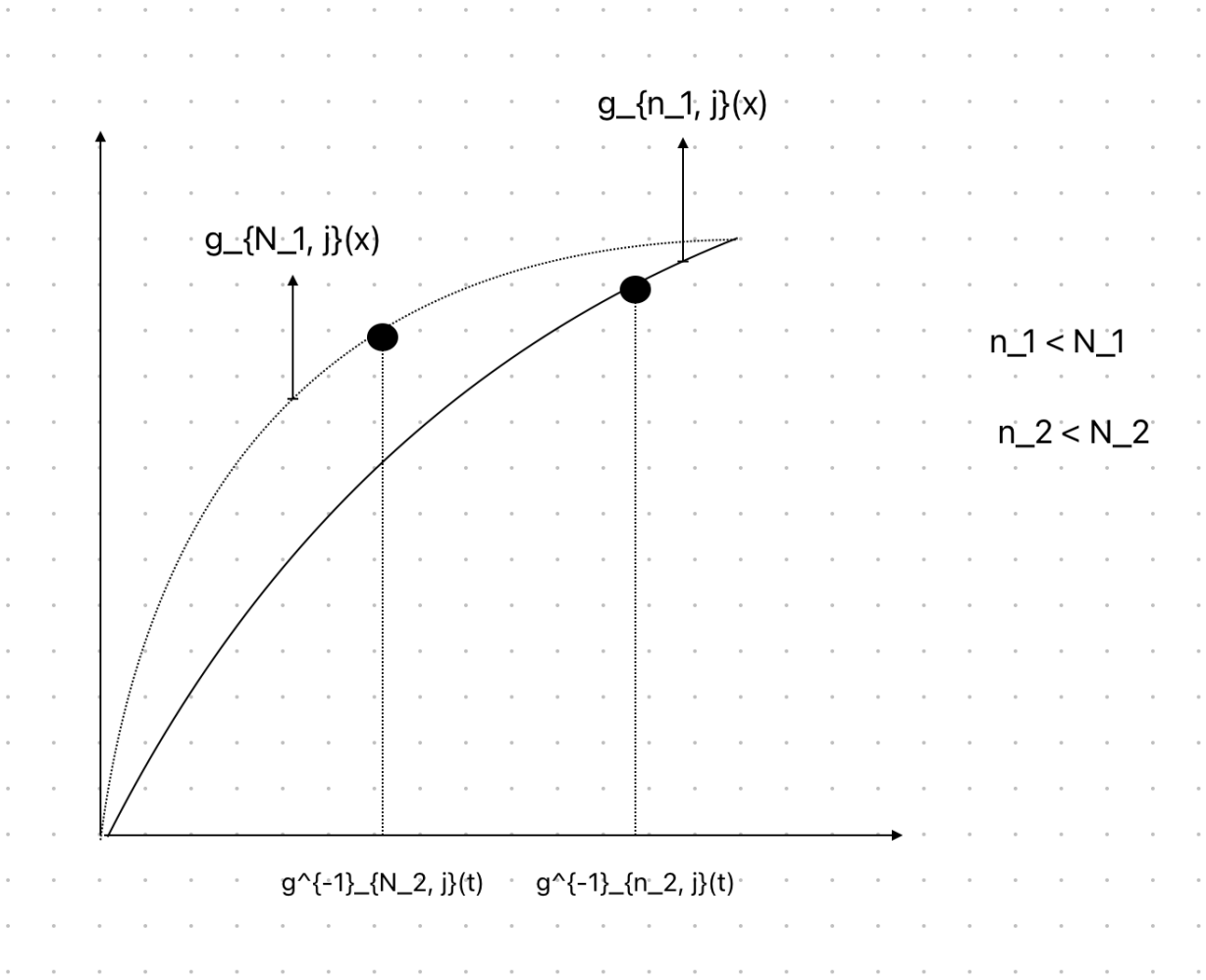
Now we have

$$g_{N_1,k}(g_{n_2,j}^{-1}(t)) \geq g_{N_1,k}(g_{N_2,j}^{-1}(t)) \tag{2}$$

Based on 1 and 3, we have

$$g_{n_1,k}(g_{n_2,j}^{-1}(t)) \geq g_{N_1,k}(g_{N_2,j}^{-1}(t)) \tag{3}$$

□


g_{n_1, j}(x)
g_{N_1, j}(x)
n_1 < N_1
n_2 < N_2
g^{-1}_{N_2, j}(t)
g^{-1}_{n_2, j}(t)